



Progesterone and Negative Emotionality Across and Between Ovulatory Cycles: A Study of Romantically Involved Women

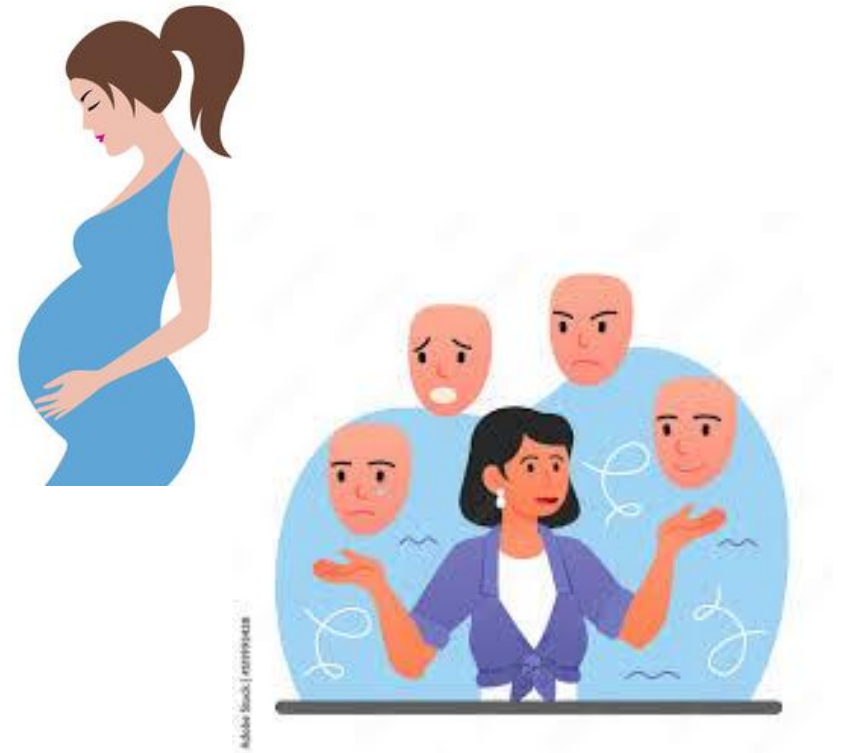
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Ovulatory Adaptations

Progesterone and Estrogen orchestrate physiological changes across the menstrual cycle.

The same hormones also facilitate psychological changes relevant to different phases of the ovulatory cycle.



Social Support Vigilance Hypothesis

(Reynolds et al., 2018)

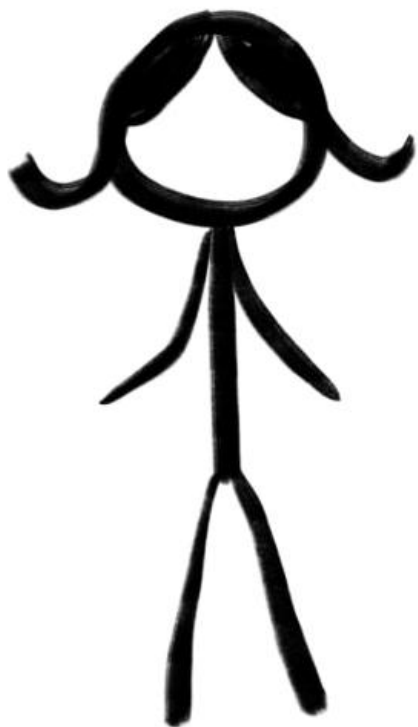
1. Rise in progesterone levels after ovulation prepares the female body for possible pregnancy
2. Pregnancy is a particularly vulnerable state for women
3. As pregnancy is a vulnerable state, women may benefit from creating a safe and supportive environment if they are pregnant
4. Thus, we may expect a positive association between progesterone and anxiety, which may promote an environment conducive to pregnancy

Social Support Vigilance Hypothesis

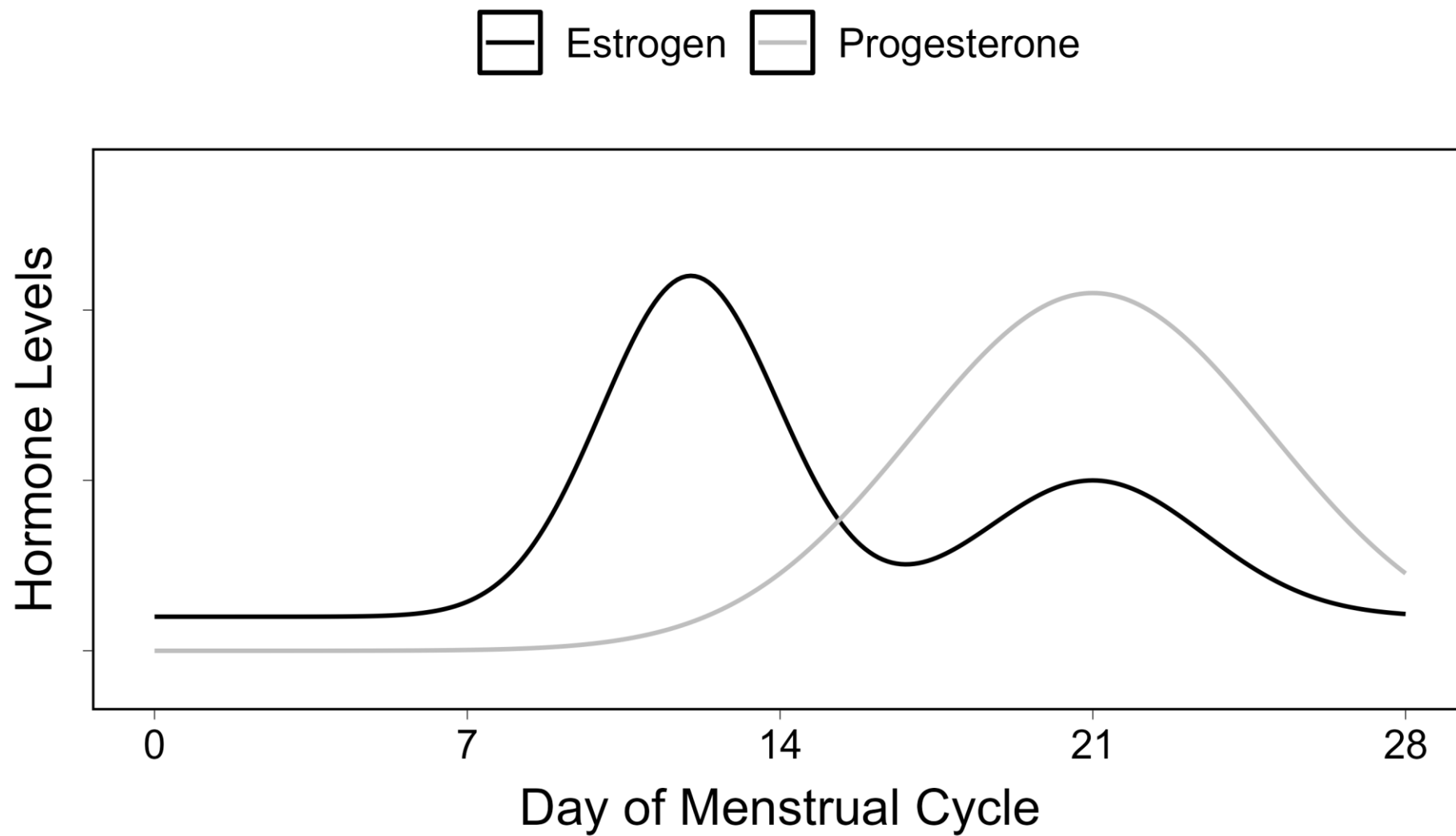
(Reynolds et al., 2018)

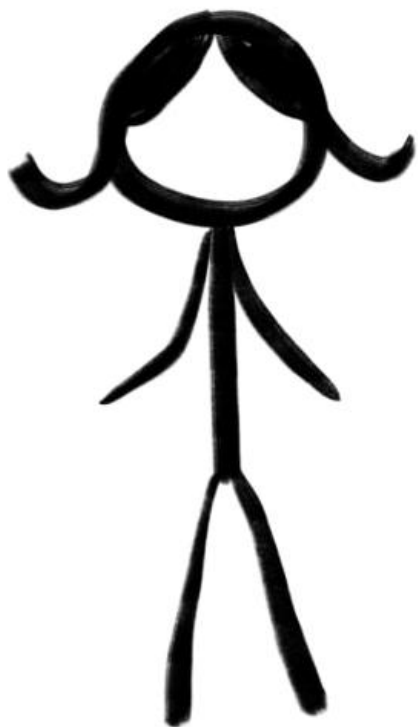
Women post-ovulation should be more concerned about the support they are receiving from their partner and others, which is driven by increasing progesterone levels



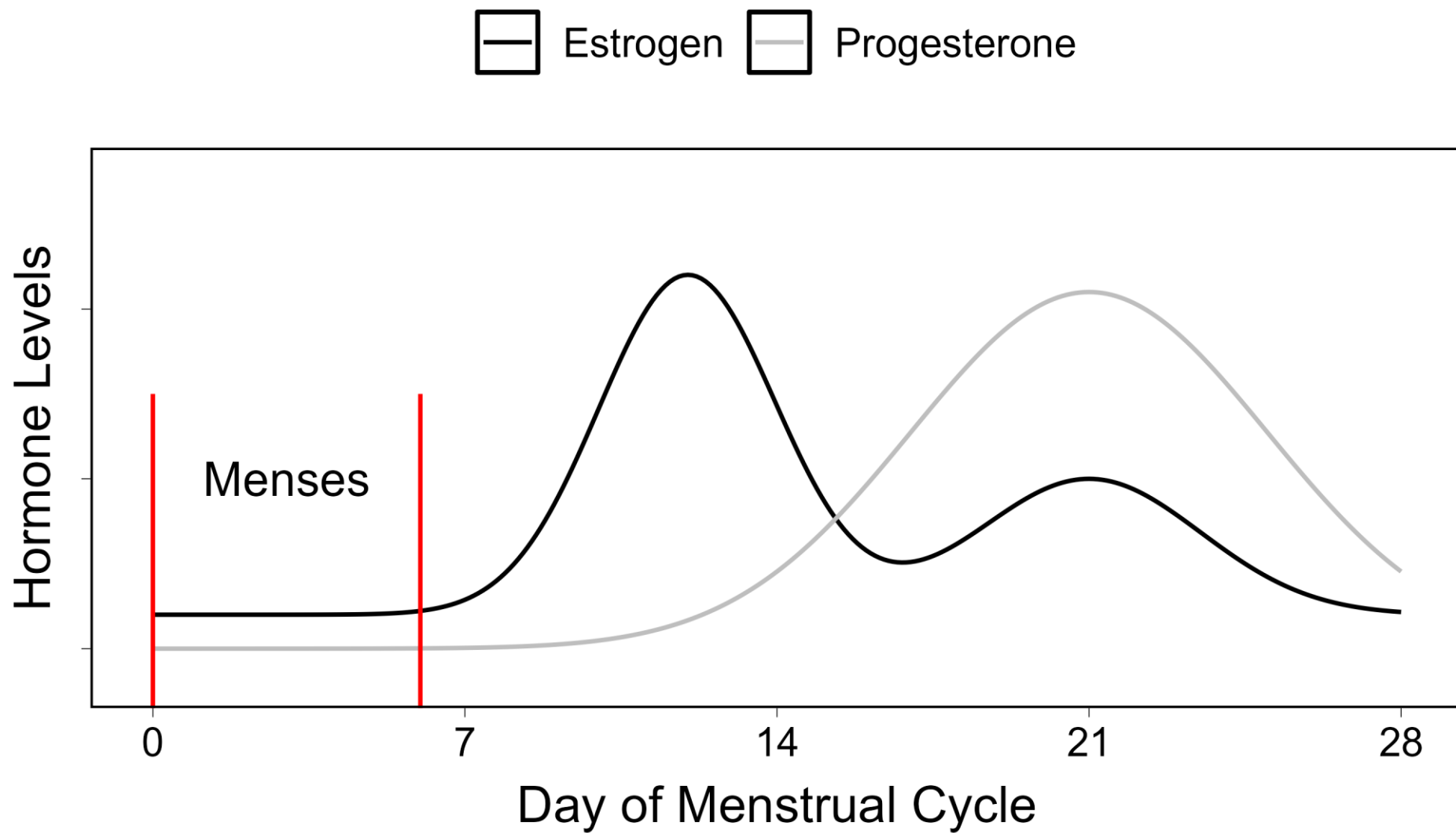


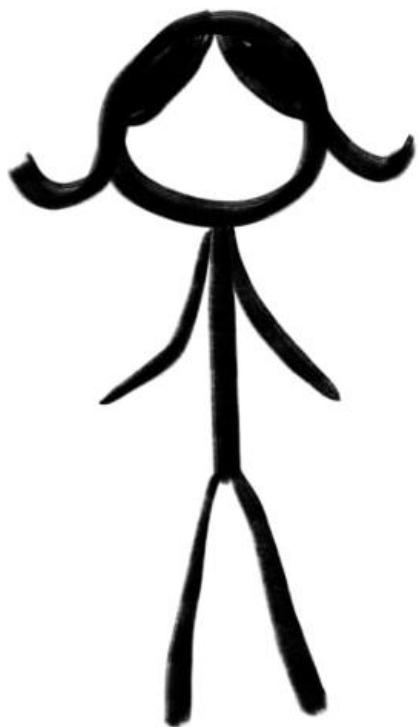
Eve



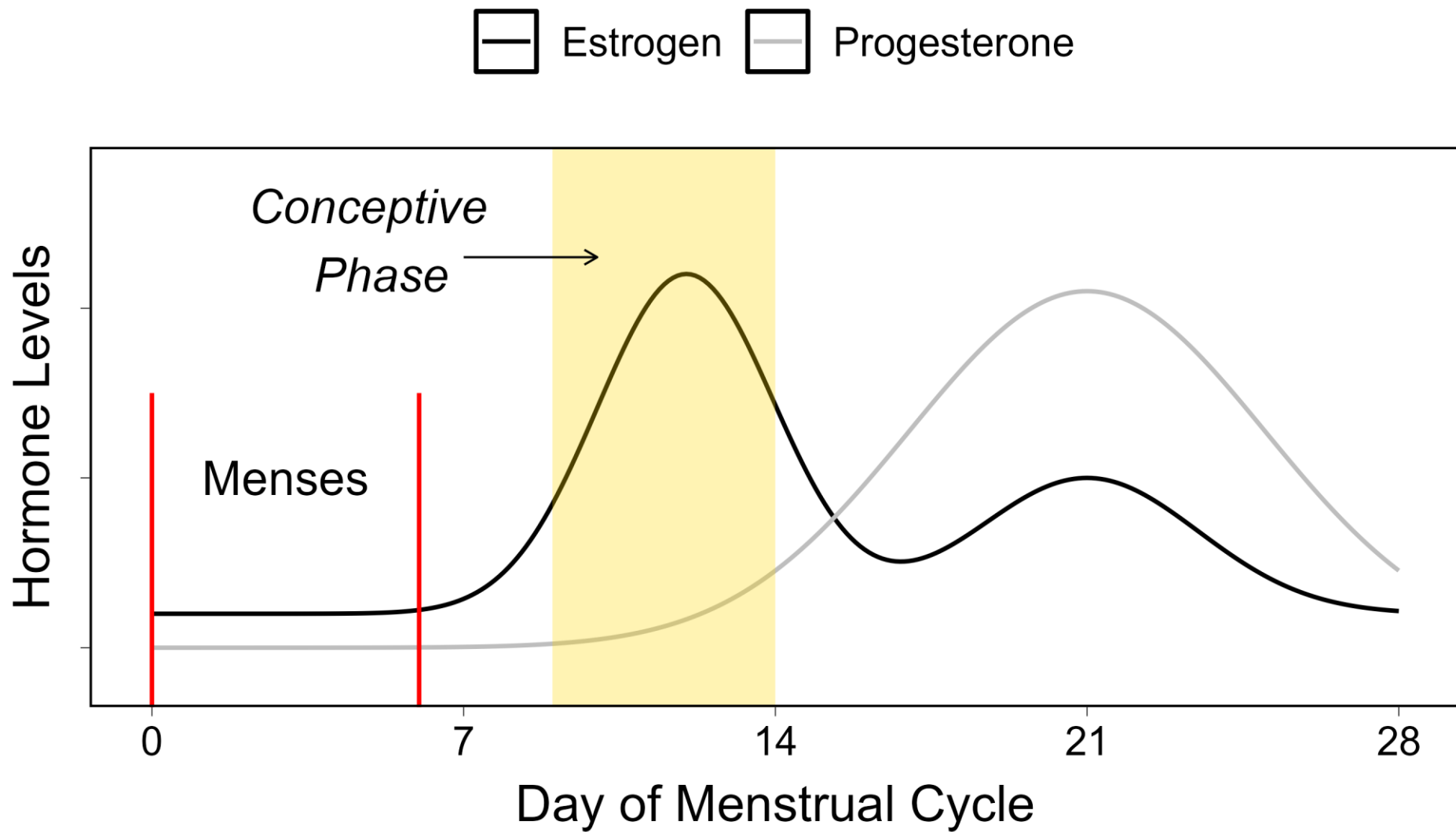


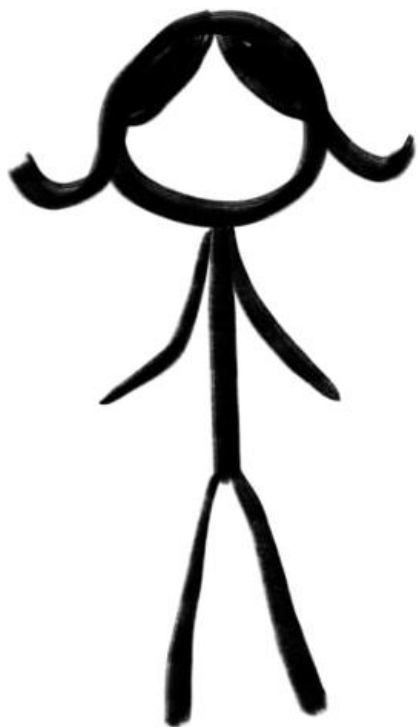
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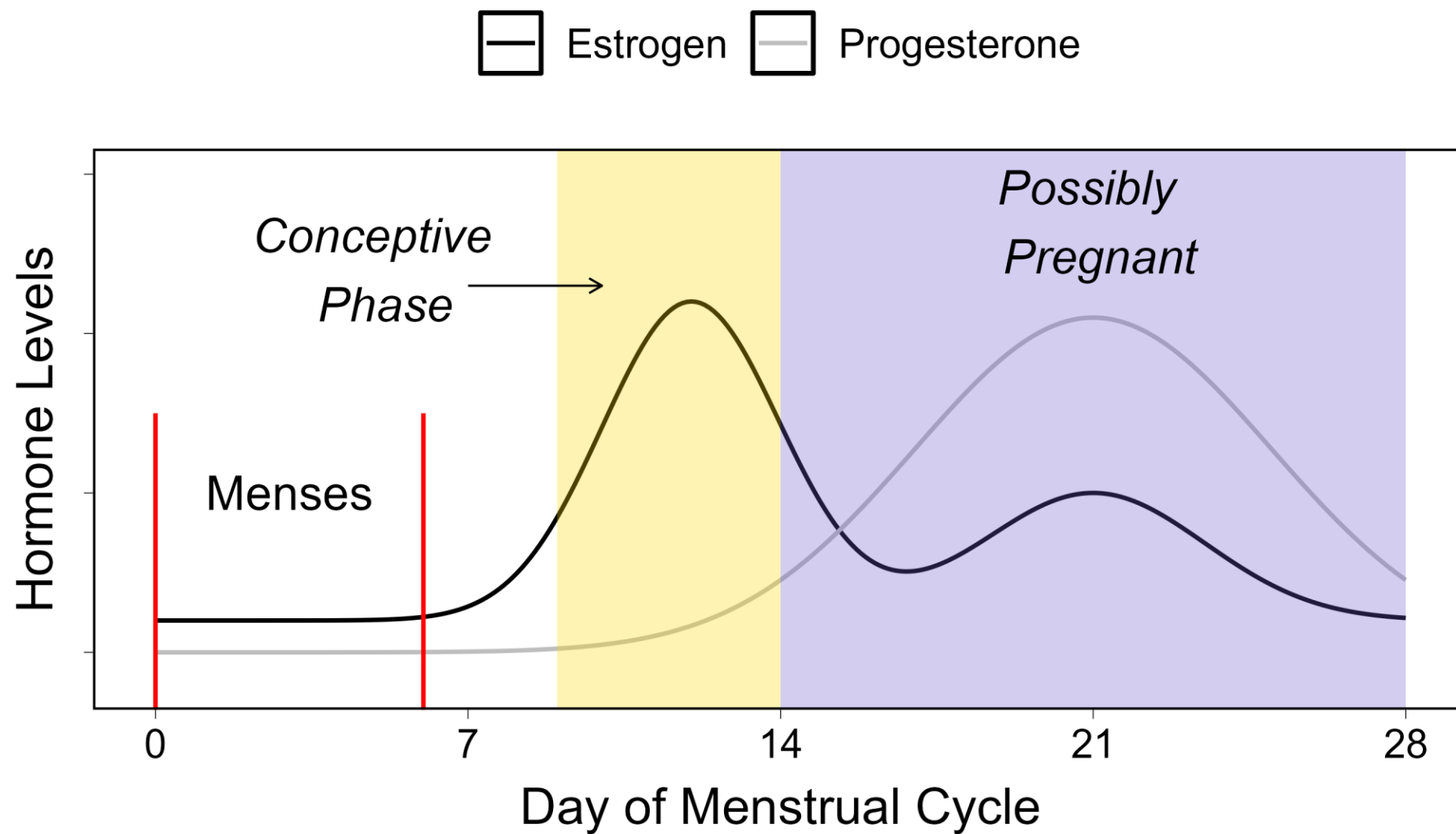


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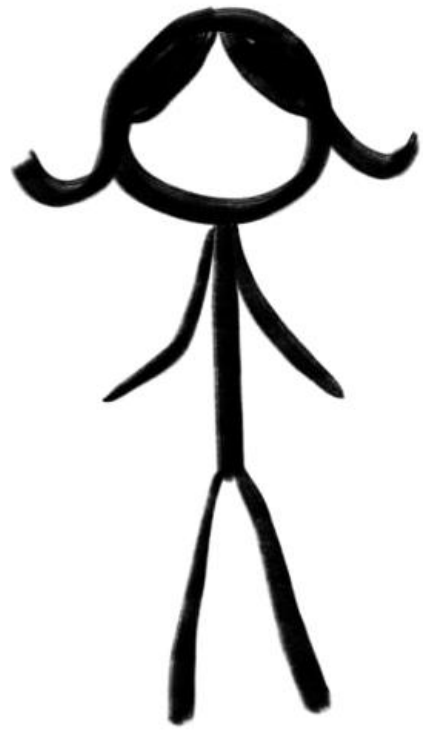




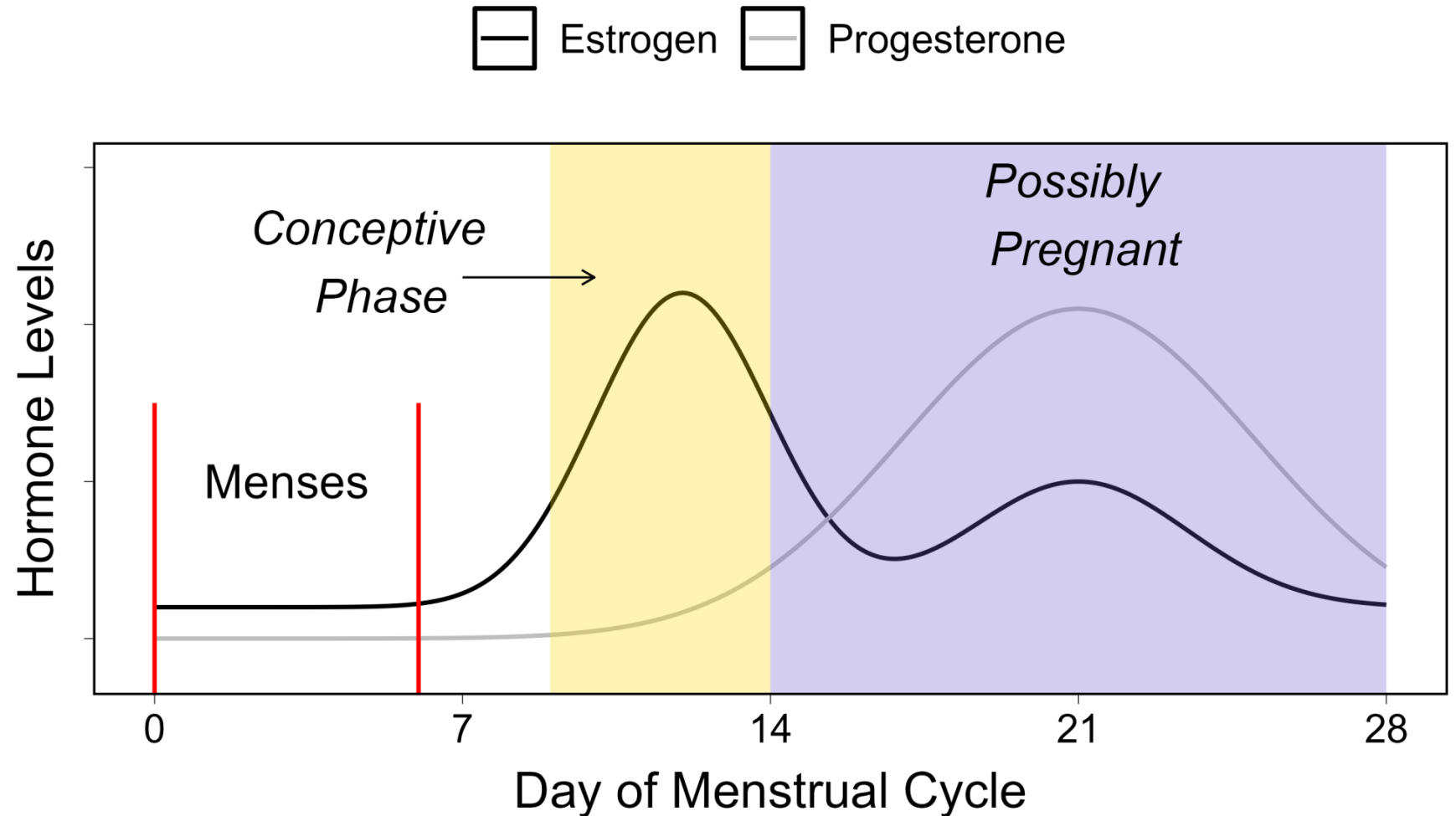
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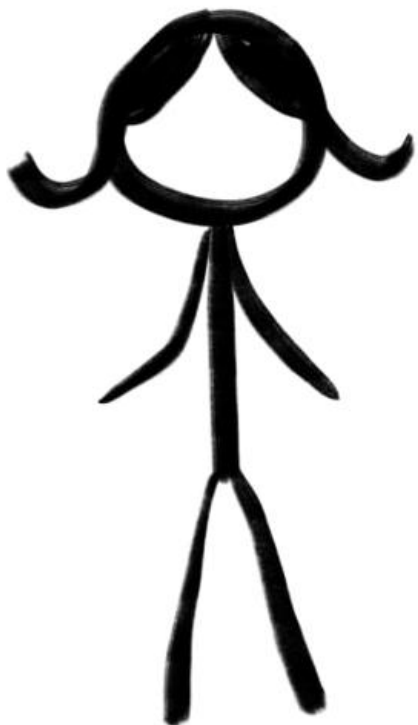


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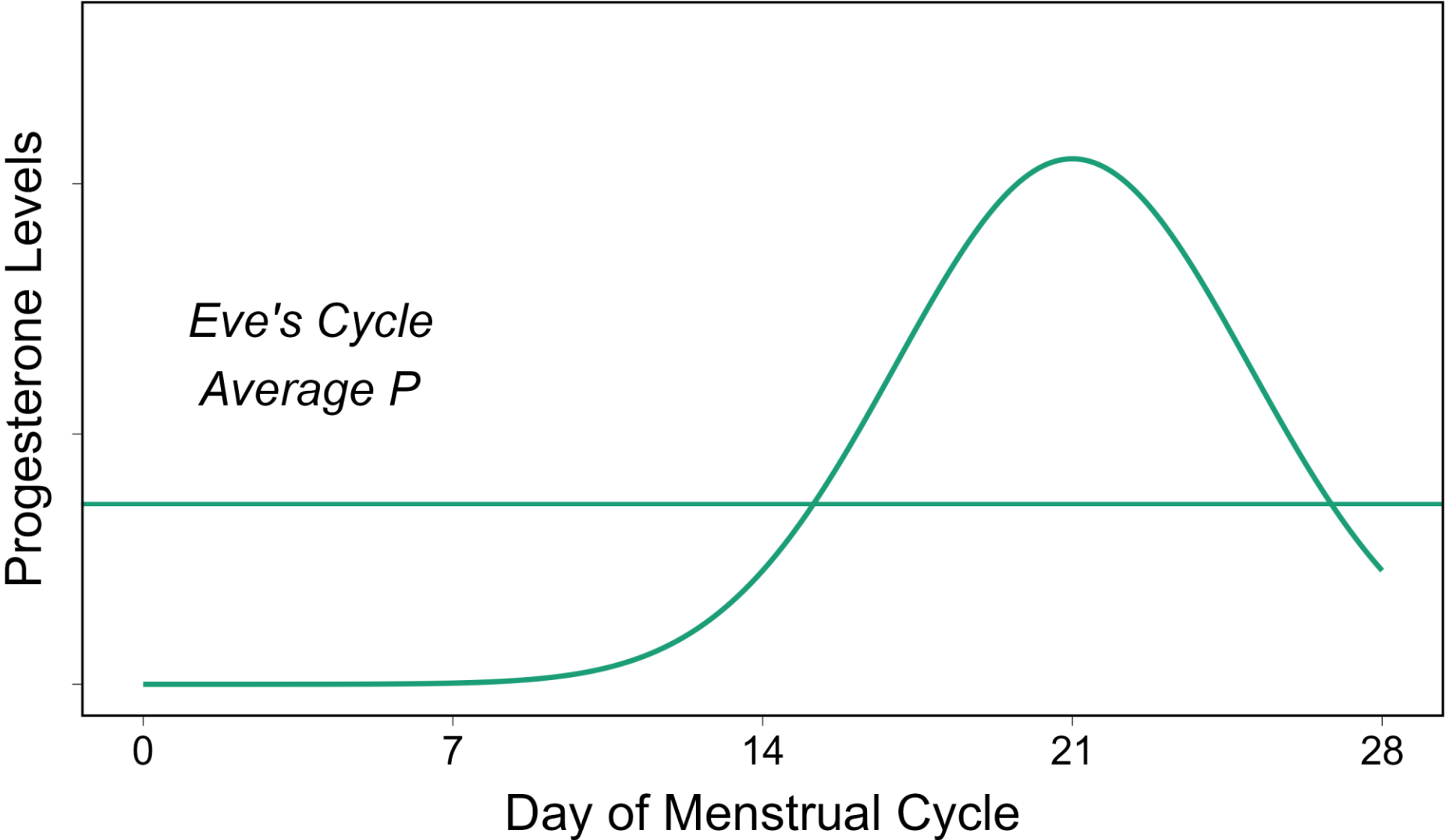


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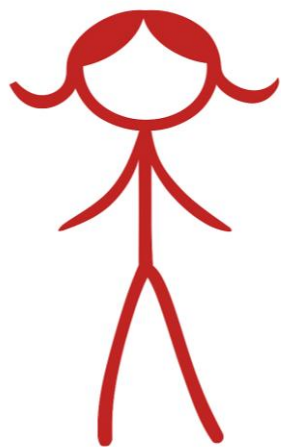




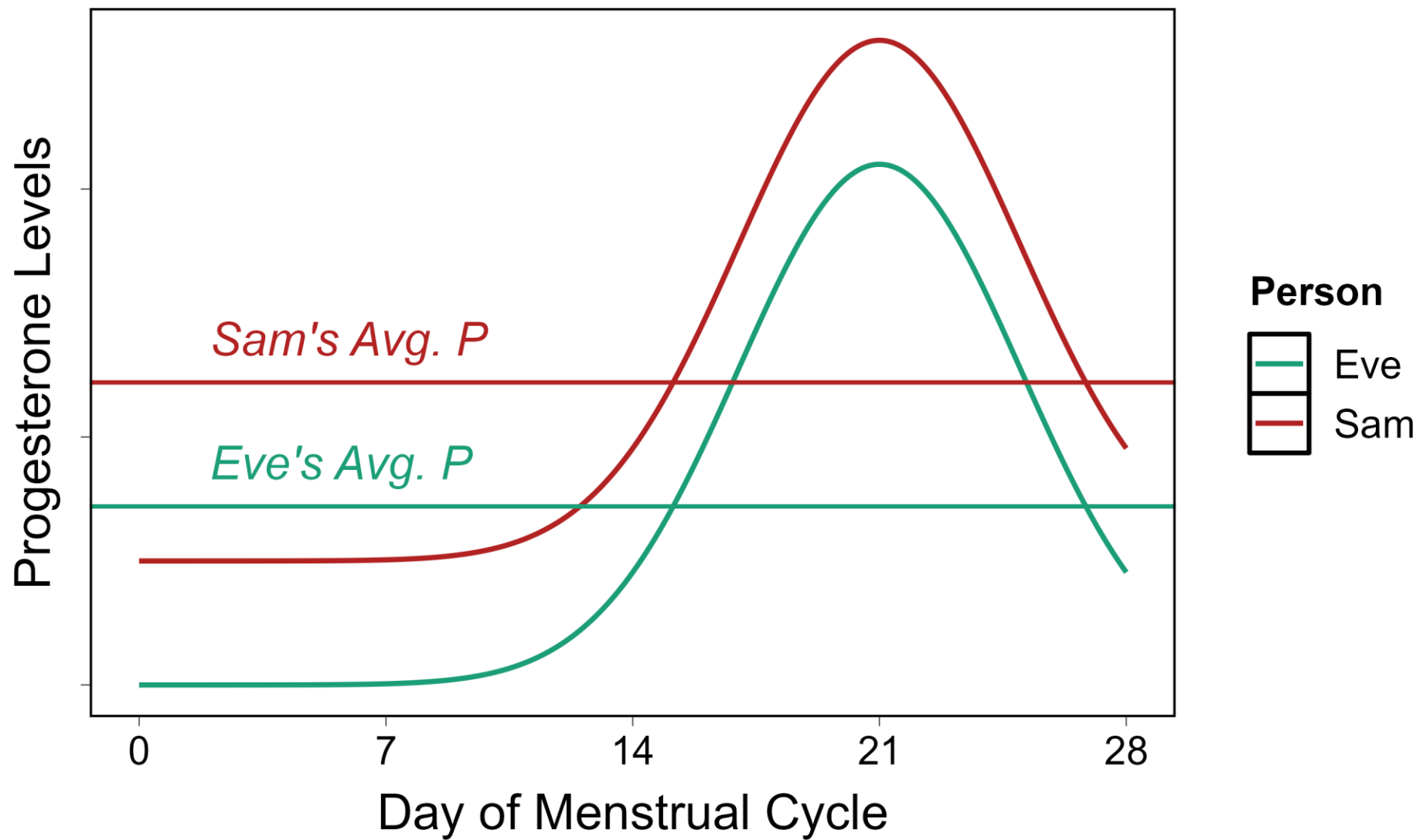
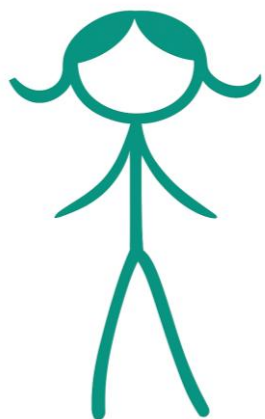
Eve



Sam



Eve



Reynolds et al. (2018) Findings

Study 1:

- Independent Variables
 - Progesterone levels
 - Estradiol levels
- Dependent Variable
 - Single item measure of anxiety: “I feel anxious today.”
- Results
 - No within-woman association
 - Positive between-women association

Study 2:

- Independent Variables
 - Progesterone levels
 - Cortisol levels
- Dependent Variable
 - Experience in Close Relationships scale (Fraley et al., 2000)
 - Attachment anxiety
- Results
 - Positive within-woman association
 - Positive between-women association

The Current Study

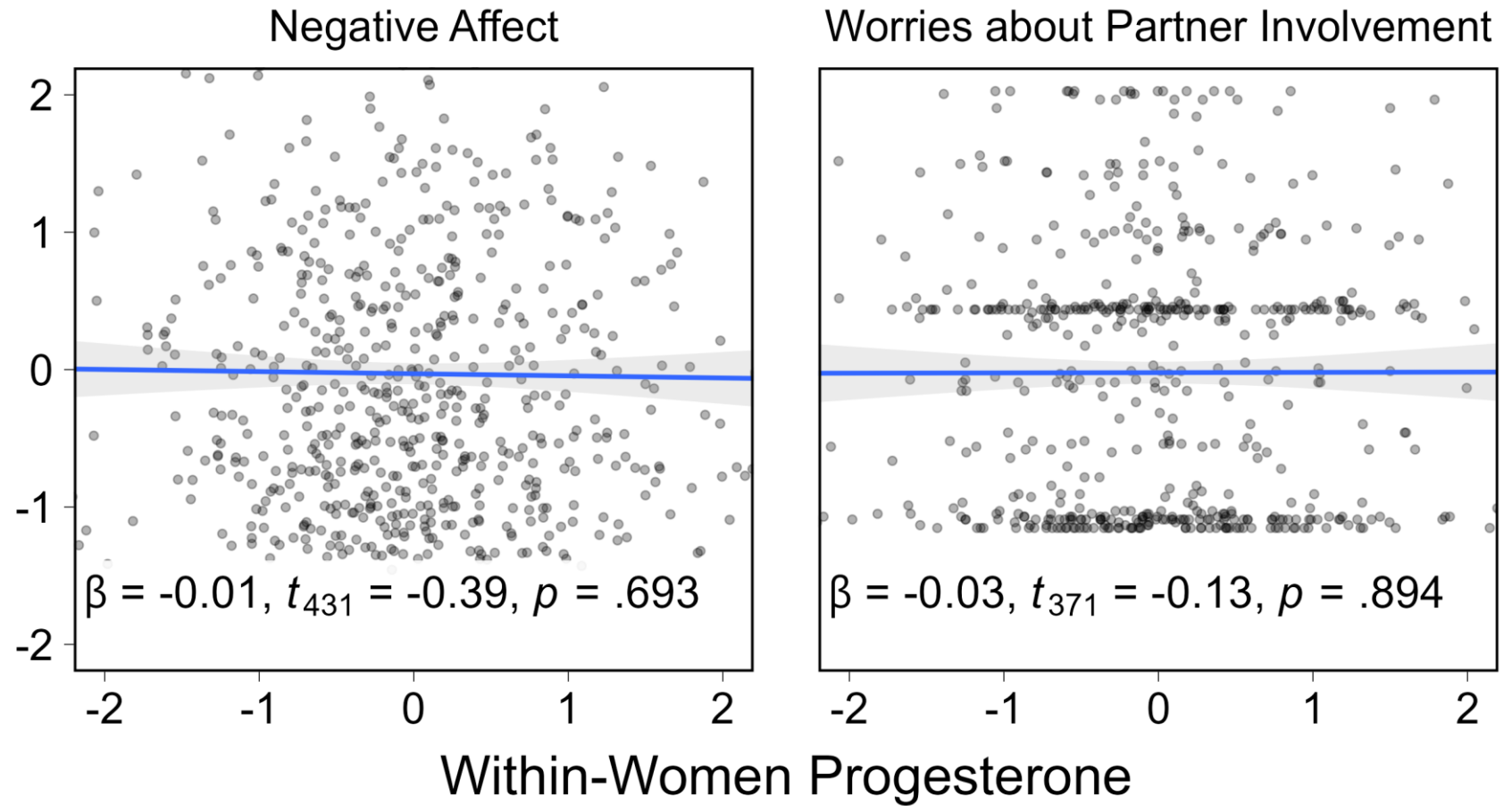
- ❖ 180 women attended between 2 and 4 lab sessions ($M = 3.6$).
- ❖ At each session...
 - ❖ Urine sample to assay hormones: progesterone, estrogen (estradiol), cortisol, and testosterone
 - ❖ Negative Emotionality measure (6 items; e.g., “How much have you felt sad in the last two days”).
 - ❖ Worries about romantic partner’s involvement (3 items; e.g., “I worried about my partner’s involvement.”)

Inclusion Criteria:

- ❖ In a heterosexual romantic relationship
 - ❖ At least 3 months long
- ❖ Age 18-35 ($M = 20.9$)
- ❖ Naturally cycling

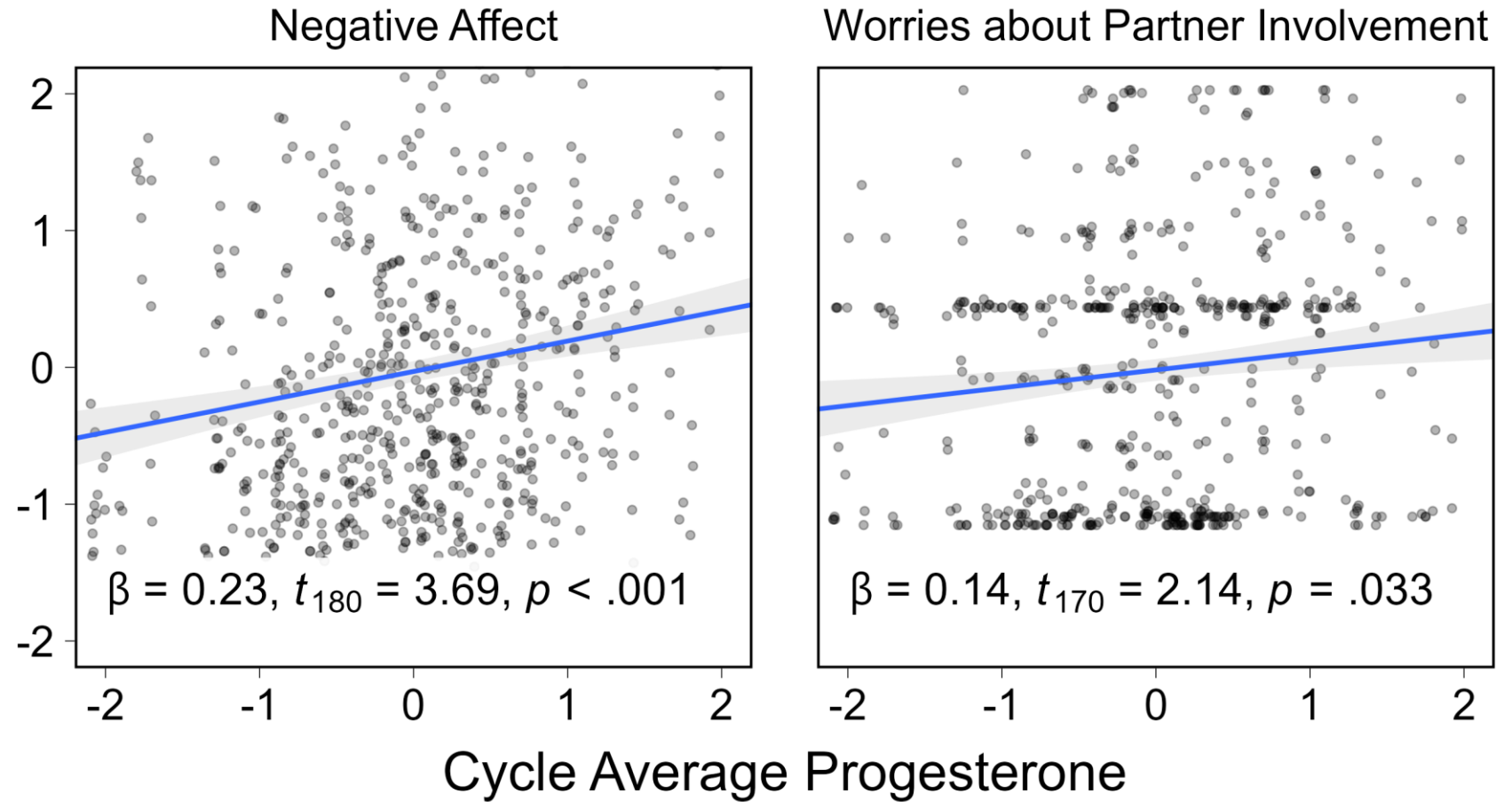
No Within-Women Effects

When “Eve” had higher P levels in her cycle she was not more likely to have more negative affect or worries about her romantic partner’s involvement.



Significant Cycle Average Effects

When Sam had a higher cycle average amount of P than Eve, Sam was also more likely to have more negative affect and worries about her romantic partner's involvement.



Confounds?

All results discussed here included controls for:

- Within and between E, C, and T
- Menses

Other analyses included additional controls:

- Weekend effect (no change)
- Age (no change)
- Individual differences (between-women effects 30% smaller)
 - Friend and Family Support
 - Big Five Personality Traits
 - Attachment Styles

Progesterone



**Worries about
Partner Involvement**

Reverse Causal Process?

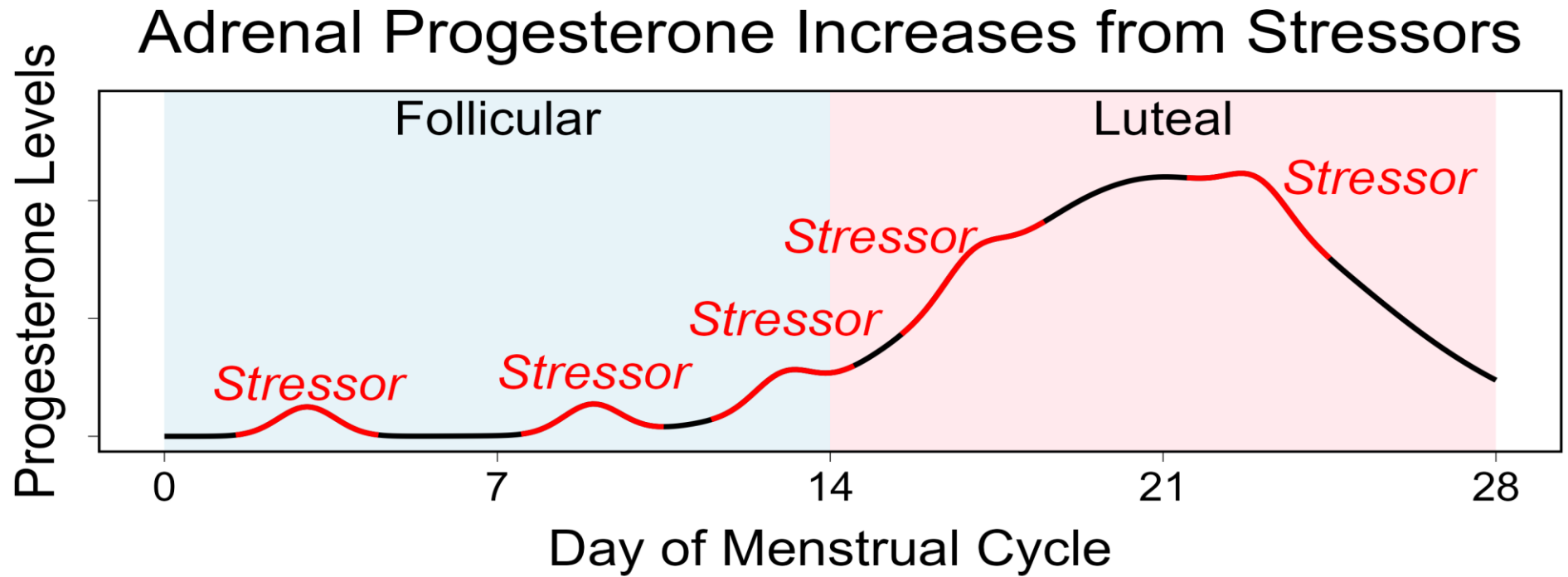
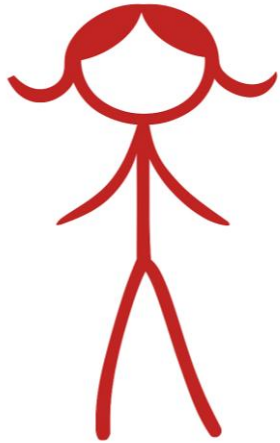


On face value, IF Stressors (Negative Emotions, Worries) cause increased Progesterone levels...

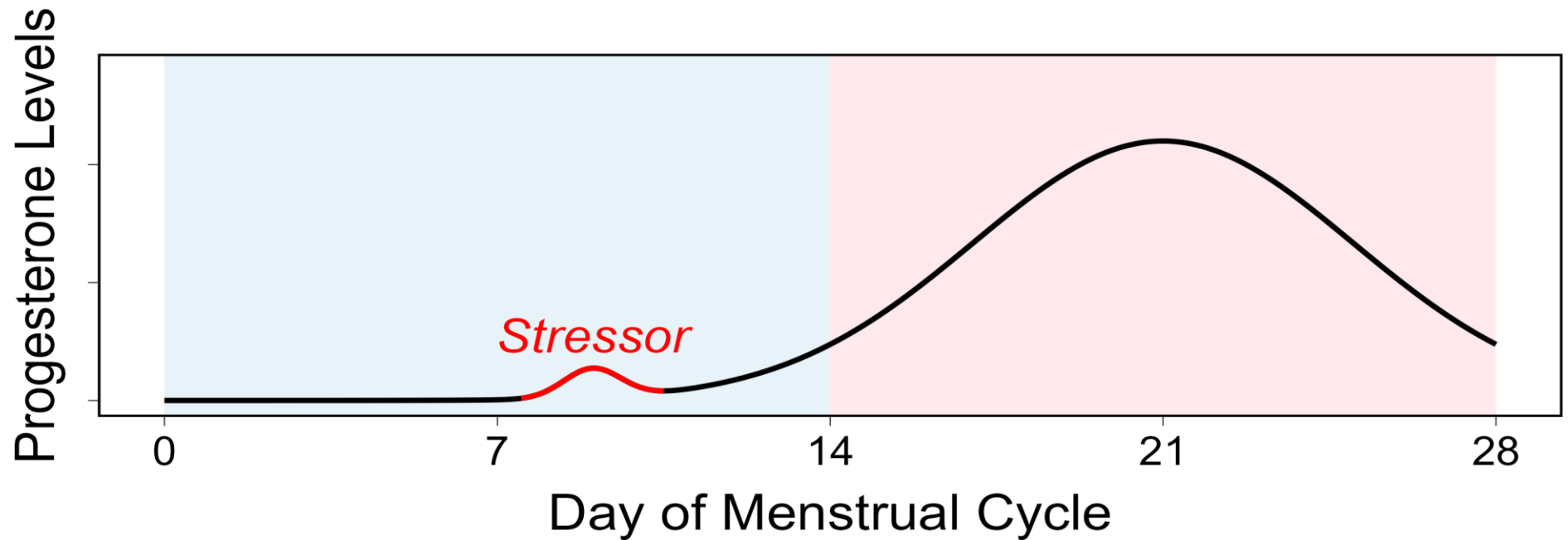
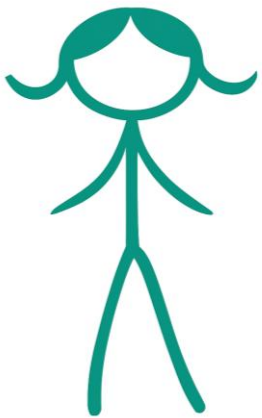
THEN Progesterone should be associated with...

- 1) the average amount of stressors across the cycle between women (significant positive effect)

Sam



Eve



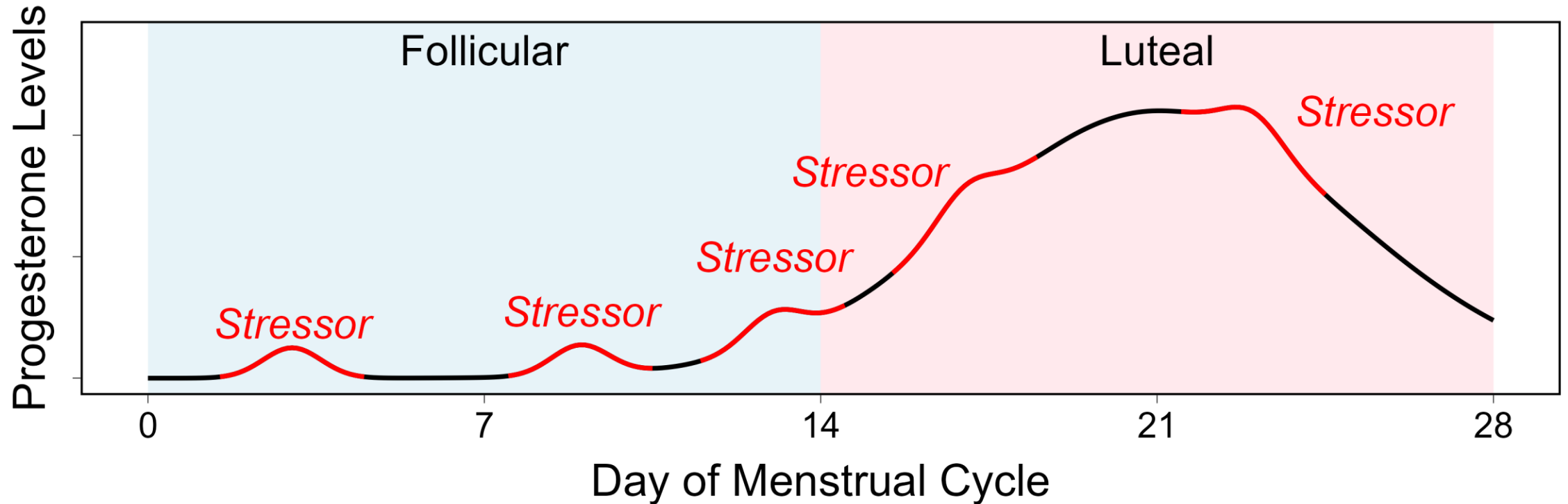
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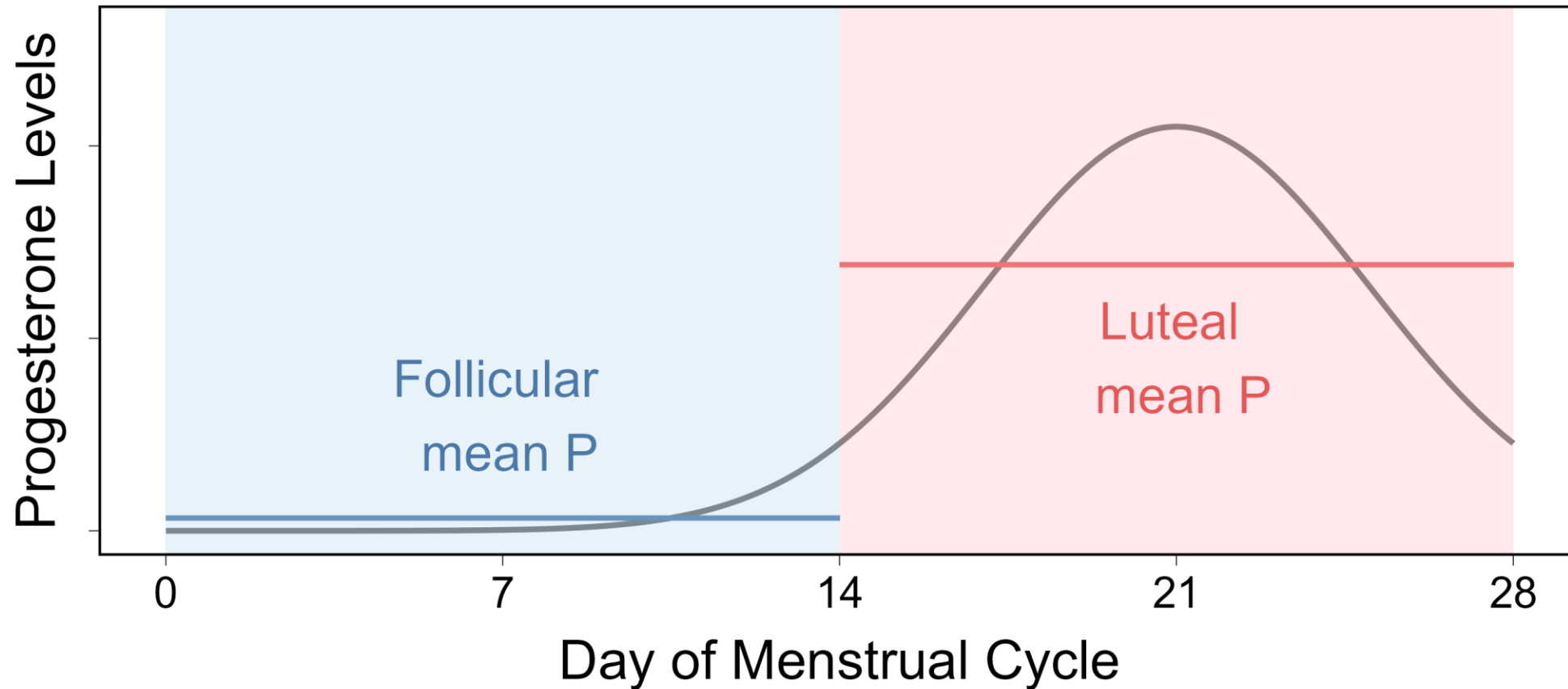
- 1) the average amount of stressors across the cycle between women (significant positive effect)
- 2) occasion specific anxious experiences within women (no effect)

However, under this causal scenario, there may be good reason to not expect the within women effect...

Ovarian Progesterone Fluctuations Greatly Exceed Adrenal Stress-Induced Spikes



If Stressors Cause Increased Progesterone, Mean Follicular Phase Levels Should Positively Predict Negative Emotionality and Worries



Progesterone Associations by Mean Follicular and Luteal Phase

Negative Emotionality

1) Follicular Phase Mean 

2) Luteal Phase Mean 

Worries about Romantic Partner Involvement

1) Follicular Phase Mean 

2) Luteal Phase Mean 

What's happening?

- Lack of within-person effect is troubling for social support vigilance hypothesis.
- Unclear what causal process is at play to generate the consistent between-woman association between progesterone and negative affect and worries about romantic partner's relationship involvement.
 - Additional analyses consistent with reverse causal process for negative affect

Manuscript and Supplemental Materials

